NextInResearch

Workshop #5: Scientific Writing + Responsible AI Usage

Guidelines • Citations + Sourcing • Introduction • Methods • Results • Discussion • Responsible AI Usage

Guidelines of Scientific Writing

- ✓ **Be clear and concise**: Use plain language + don't add filler
- ✓ **Use appropriate tenses**: Past tense for your methodology and findings; present tense for established facts and conclusions
- ✓ **Write for your audience**: Read recent papers in your target journal to match its style and terminology

Citations & Sourcing

- Most journals accept a **references page** at the end of your article, with numbers and full citations
- **In-text citations**: use numbers in brackets — [1…3], [1,5], [6] (the convention for many larger journals)
- Numbers correspond to the order listed in the References Page, normally numbered **in order of usage**

Parts of the Introduction

1. Hook and Background

Grab the reader's attention, explain why the topic matters, and define key terms that may not be commonly known

2. Literature Review

Show what is already known and establish credibility with a brief summary of key, recent, relevant studies. Treat your sources as a **collection!**

3. Research Gaps + Your Why

Justify why your paper is needed: a specific limitation, contradiction, or missing piece. **Ex**: "Although previous studies have examined [Topic X], Topic Y remains understudied"

4. Objective

Tell the reader exactly what your study will do. **Ex**: "The present study aims to ___." "We/I hypothesized that ___"

A Paper at a Glance

1. Introduction → 2. Materials and Methods → 3. Results → 4. Discussion → References

Writing the Body: Methods, Results, Discussion

2. Materials + Methods

Split into **chronological subsections** (2.1, 2.2, …) so the paper is easy to read and **replicate**.

2.1 Study Design: overarching framework (survey, randomized controlled trial, computational, in-vitro, in-vivo…)

2.2 Participants/Materials: ALL the info someone would need, incl. inclusion/exclusion criteria (n=45)

2.3 Procedures and Data Analysis: chronological account of experiments + statistical tests and programs used

3. Results

Include: data summary (text, tables, figures), significant AND non-significant outcomes, exact p-values; connect back to your research questions

Avoid: interpreting findings (yet!) and dumping unanalyzed raw data — **SUMMARIZE** it

Ex: "Grade 6 middle schoolers exercised significantly more than grade 8 (p=0.002)"

4. Discussion

Interpret your results: give a possible answer for WHY — "This result SUGGESTS ___"

Compare with existing literature: how does your result agree/disagree with others?

Acknowledge study limitations

Key Reminders

Save interpretation for the Discussion — the Results section only reports.

Don't mix the order of methods and results.

Number subsections by section: Methods = 2.1, 2.2…; Results = 3.1…

Responsible AI Usage

LLMs (ChatGPT, Claude, Gemini): AI systems trained on mass amounts of data that answer questions, write code, and solve problems by predicting the most statistically probable next words.

Agents: smart helpers that can think, decide, and do things to achieve a goal — e.g. building an app: one agent builds the front-end, one sets up the database, one tests code.

Use AI to GROW, not STUNT your learning